PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	28 October 2026	Maximum Marks	2 Marks
Team ID	EN25CS3011166	Project Name	Health360: Cardiovascular Risk Analytics

Define Your Technology Stack:
Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze healthcare metrics.	**Python 3.11, SQL, HTML5, CSS3**	Python enables metric manipulation in Pandas; SQL computes AVG(Heart Disease) and risk counts in Snowflake.
2	**Database / Data Warehouse**	Warehouse used to store and manage structured patient health data for analytics.	**Snowflake / MySQL / Excel / CSV (smart_healthcare_dataset)**	Centralized data warehouse to process patient risk indicators (BMI, Glucose, Blood Pressure, Fatigue).
3	**Data Ingestion**	Used to load source data files or live patient feeds into the analytical environment.	**Snowflake Stages / COPY INTO / Tableau Connections**	Automates bulk loading of clinical dataset records directly into memory for interactive dashboard updates.
4	**Data Transformation & Business Logic**	Logic to clean, join, aggregate data and compute risk scores and categorical bins.	**Tableau Prep, Snowflake SQL, Tableau Calculated Fields**	Bins Blood Pressure, BMI, and computes AVG(Heart Disease Risk) across age groups 19-79.
5	**Data Visualization / Analytics**	Used to create interactive heatmaps, line graphs, bar charts, and stroke/cardiac risk panels.	**Tableau Public, Tableau Desktop, Python (Seaborn)**	Generates visualizations like "Fatigue v/s Stroke Risk", "Exercise Level vs Risk", and "Patient Risk Heatmap".
6	**Front-End / User Interface**	Interface through which users interact with filters, parameters, and health risk dashboards.	**Tableau Public Web Embed, Streamlit / Flask**	Provides clinicians and administrators with dynamic filters for Age, Chest Pain, Dizziness, and BMI.